

PRETENOL C TABLET

DESCRIPTION: PRETENOL C 100 TABLET: White to off-white, 10 mm round, biconvex, scored on one side and 'C' marking on the other side, tablet
PRETENOL C 50 TABLET: White to off-white, 8 mm round, biconvex, 'C' marking on one side and plain on the other side, tablet

COMPOSITION: PRETENOL C 100 TABLET: Each tablet contains Atenolol 100 mg, Chlorthalidone 25 mg
PRETENOL C 50 TABLET: Each tablet contains Atenolol 50 mg, Chlorthalidone 12.5 mg

PHARMACODYNAMICS: Atenolol and Chlorthalidone have been used singly and concomitantly for the treatment of hypertension. The antihypertensive effects of these agents are additive and studies have shown that there is no significant interference with bioavailability when these agents are given together in the single combination tablet. In patients with more severe hypertension Pretenol-C may be administered with other antihypertensive such as vasodilators.

Atenolol: A beta-adrenoreceptor blocking drug which acts preferentially on beta-receptors in the heart. Selectivity decreases with increasing dose. It has little intrinsic sympathomimetic activity and no membrane stabilising activity. Atenolol is a racemic mixture and its activity resides in the S(-) enantiomer. It reduces raised blood pressure by an unknown mechanism and also inhibits exercise induced tachycardia and decreases plasma renin concentration. It causes slight airways obstruction but less than that seen with non-selective beta-blockers. The inhibition of exercise induced tachycardia is correlated with blood levels, but there is no correlation between plasma concentrations and antihypertensive effect. Atenolol is effective and well-tolerated in most ethnic populations although the response may be less Afro-Caribbean black patients.

The possible mechanism of the anti-anginal activity of Atenolol appears to be due to a reduction in left ventricular work and oxygen utilisation resulting (mainly) from the decrease in heart rate and contractility. The anti-arrhythmic effect of Atenolol is apparently due to its anti-sympathetic effect. There is no evidence that membrane stabilising activity or intrinsic sympathomimetic activity are necessary for anti-arrhythmic efficacy. By its anti-sympathetic effect, Atenolol depresses sinus node function, atrioventricular node function and prolongs atrial refractory periods. It has no direct effect on electrophysiological properties of the His-Purkinje system. Because of their negative inotropic effects, beta-adrenoreceptor blocking agents should be avoided in uncontrolled heart failure.

Chlorthalidone: A monosulfonamide diuretic which differs chemically from thiazide diuretics in that a double ring system is incorporated in its structure. It is an oral diuretic with prolonged action and low toxicity. The diuretic effect of the drug occurs within 2 hours of an oral dose and continues for up to 72 hours. It produces copious diuresis with greatly increased excretion of sodium and chloride. At maximal therapeutic dosage chlorthalidone is approximately equal in its diuretic effect to comparable maximal therapeutic doses of benzothiazine diuretics. The site of action appears to be the cortical diluting segment of the ascending limb of Henle's loop nephron.

PHARMACOKINETICS: **Atenolol:** Although absorption of Atenolol is variable and incomplete (40-60%) the virtual lack of liver metabolism results in a relatively consistent systemic bioavailability compared to other beta-blockers. Blood levels in humans peak two to four hours after a single 100 mg oral dose and are of the order of 0.4 to 0.9 microgram / mL. Blood levels are consistent and the levels after chronic oral administration are in good agreement with those predicted from single dose results. The drug is distributed throughout the body tissues and less than 10% of the drug is metabolized, the minor urinary metabolite identified is a hydroxylated derivative. The main route of elimination is renal excretion. The plasma half-life, measured by blood level decay or urinary build-up is approximately 7 to 9 hours. In patients with impaired renal function, there is a progressive prolongation of the half-life. In patients with normal renal function, the therapeutic effect (that is, control of raised blood pressure) lasts for at least 24 hours following a 50 mg oral dose.

Chlorthalidone: **Absorption and plasma concentration:** The bioavailability of an oral dose of 50mg chlorthalidone is approximately 64%, peak blood concentrations being attained after 8 to 12 hours. For doses of 25 and 50 mg, C_{max} values average 1.5 mcg/mL (4.4 micromol/L) and 3.2 mcg/mL (9.4 micromol/L) respectively. For doses up to 100 mg there is a proportional increase in AUC. On repeated daily doses of 50mg, steady-state blood concentrations, measured at the end of the 24-hour dosage interval, averaging 7.2 mcg/mL (21.2 micromol/L) are reached after 1 to 2 weeks. **Distribution:** In blood, only a small fraction of chlorthalidone is free, due to extensive accumulation in erythrocytes and binding to plasma proteins. Owing to the large degree of high-affinity binding to the carbonic anhydrase of erythrocytes, only some 1.4% of the total amount of chlorthalidone in whole blood was found in plasma at steady state during treatment with 50 mg doses. In vitro, plasma protein binding of chlorthalidone is about 76%, and the major binding protein is albumin. Chlorthalidone crosses the placental barrier and passes into breast milk. In mothers treated with 50 mg chlorthalidone daily before and after delivery chlorthalidone levels in foetal whole blood are about 15% of those found in maternal blood. Chlorthalidone concentrations in amniotic fluid and in the maternal milk are approximately 4% of the corresponding maternal blood level. **Metabolism and elimination:** Chlorthalidone is eliminated from whole blood and plasma with an elimination half-life averaging 50 hours. The elimination half-life is unaltered after chronic administration. The major part of an absorbed dose of chlorthalidone is excreted by the kidneys, with a mean renal plasma clearance of 60 mL/min. Metabolism and hepatic excretion into bile constitute a minor way of elimination. Within 120 hours, about 70% of the dose is excreted in the urine and in the faeces mainly in unchanged form. **Special patient groups:** Renal dysfunction does not seem to alter the pharmacokinetics of chlorthalidone, the rate-limiting factor in the elimination of the drug from blood or plasma being most probably the affinity of the drug to the carbonic anhydrase of erythrocytes. In elderly patients, the elimination of chlorthalidone is slower than in healthy young adults, although absorption is the same. Therefore, close medical observation is indicated when treating patients of advanced age with chlorthalidone.

INDICATIONS: Treatment of hypertension in patients for whom the effect of treatment with either beta-blockers or diuretic alone is inadequate. The patient should be stabilised on the two drugs individually before transfer to the fixed combination if the dosage is appropriate.

RECOMMENDED DOSAGE: Dosage of individual drugs to be titrated and patients to be stabilized on the two drugs individually before transfer to the fixed combination if dosage is appropriate. A dose of 50/12.5mg daily of Pretenol C should prove effective in most cases. If the response is considered inadequate then a dose of 100/25mg daily should be tried. However, higher doses offer a significant risk of hypokalaemia and periodic monitoring of potassium levels should always be undertaken in these patients. Above this dose level there is unlikely to be any further fall in blood pressure and another anti-hypertensive drug such as a vasodilator should be added.

Children: There is no paediatric experience with Pretenol-C therefore this preparation is not recommended for children.

A daily dose of 100/25mg may be inappropriate in the older patient. For these patients, a dose of 50/12.5mg daily is recommended. The addition of a small dose of a third agent, eg. a vasodilator, may be appropriate should hypertensive control not be achieved on a dose of 50/12.5mg daily alone. In patients with severe renal impairment a reduction in daily dose or in frequency of administration may be necessary.

Pretenol-C 100 Tablet (100mg Atenolol/25mg Chlorthalidone) is scored and can be halved to provide two doses of 50mg Atenolol/12.5mg Chlorthalidone each. The recommended dosing for the 50mg/12.5mg strength is the same as that for Pretenol-C 50 Tablet.

ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS:

1. Bronchospasm: Beta adrenergic blockade of the smooth muscle of bronchi and bronchioles may result in an increased airways resistance. These drugs also reduce the effectiveness of asthma treatment. This may be dangerous in susceptible patients. Therefore, beta blockers are contraindicated in any patients with a history of airways obstruction or tendency to bronchospasm. Use of cardioselective beta-blockers can also result in severe bronchospasm. If such therapy must be used, great caution should be exercised. Alternative therapy should be considered.
2. Congestive heart failure.
3. Allergic disorders (including allergic rhinitis) which may suggest a predisposition to bronchospasm.
4. Right ventricular failure secondary to pulmonary hypertension.
5. Significant right ventricular hypertrophy.
6. Sick sinus syndromes.
7. Sinus bradycardia (less than 45-50 beats/minute)
8. Second and third degree A-V block
9. Shock (including cardiogenic and hypovolaemic shock)
10. Anaesthesia with agents that produce myocardial depression (eg. ether, chloroform, cyclopropane)
11. Hypersensitivity to the drug or other sulphonamide-derived drugs
12. Hypotension
13. Metabolic acidosis
14. Severe peripheral arterial circulatory disturbances
15. Untreated pheochromocytoma
16. Pregnancy and Lactation (see Warning: Use in Pregnancy and Use in Lactation)
17. Hypertension during pregnancy
18. Creatinine clearance lower than 30mL/min
19. Conditions involving enhanced potassium loss, for example, salt-losing nephropathies and prerenal (cardiogenic) impairment of kidney function.

WARNING & PRECAUTIONS:

Warnings:

1. **Cardiac Failure:** Beta-blockade depresses myocardial contractility and may precipitate cardiac failure in some patients with a history of cardiac failure, chronic myocardial insufficiency or unsuspected cardiomyopathy, as may occur in chronic alcoholism. In patients without a history of cardiac failure, continuing depression of the myocardium may lead to cardiac failure. If signs of cardiac failure present, the patients should be fully digitalised and/or given an ACE inhibitor or vasodilators with or without a diuretic and carefully monitored. If cardiac failure persists, the beta-blocker should be withdrawn. (See Warnings 2.)

(Note: Although congestive heart failure has been considered to be a contra-indication to the use of beta-blockers, there is a growing literature on the experimental use of beta-adrenergic blocking drugs in heart failure.

As further trials are needed to identify which patients are most likely to respond to which drugs, beta-blockers should not normally be prescribed for heart failure outside of specialist centres).

2. **Abrupt Withdrawal of Therapy:** Care should be taken if beta-blockers have to be discontinued abruptly in patients with coronary artery disease. Severe exacerbation of angina and precipitation of myocardial infarction and ventricular arrhythmias have occurred following abrupt discontinuation of beta-blockade in patients with ischaemic heart disease. Therefore it is recommended that the dosage be reduced gradually over a period of about 8 to 14 days during which time the patient's progress should be reassessed. The drug may be reinstituted temporarily if the angina worsens. If the drug must be withdrawn abruptly, close observation is required. In the peri-operative period beta-blockers, should not be withdrawn, unless indicated.

3. **History of Anaphylactic Reaction:** While taking beta-adrenoreceptor blocking drugs, patients with a history of anaphylactic reaction to a variety of allergens may have a more severe reaction on repeated challenge. Such patients may be unresponsive to the usual doses of adrenaline used to treat the allergic reactions.

4. **Concomitant Therapy with Calcium Antagonists:** The concomitant use of beta blockers and calcium antagonists with myocardial depressant and sinus node activity (e.g. verapamil and, to a lesser extent diltiazem) may cause hypotension, bradycardia and asystole. Extreme caution is required if these drugs have to be used together. The dihydropyridine calcium antagonists (e.g. nifedipine) have a weaker myocardial depressant effect and can be administered cautiously with beta-blockers. If excessive hypotension develops, the calcium antagonist should be stopped or the dosage reduced.

5. **First Degree heart block:** Due to its negative effect on conduction time, caution must be exercised if Atenolol is given to patients with first degree heart block.

6. **Anti-arrhythmic Drugs:** Care should be taken when prescribing beta-blockers with anti-arrhythmic drugs. Interactions have been reported during concomitant beta-blocker therapy with the Class IA agents disopyramide, and less frequently quinidine; Class IB agents, tocainide, mexiletine and lignocaine; Class IC agents, flecainide and propafenone (not available in Australia); the Class III agent, amiodarone; and the Class IV anti-arrhythmic agents.

7. **Peripheral Circulation:** Beta-blockade may impair the peripheral circulation and exacerbate the symptoms of peripheral vascular disease.

8. **Prinzmetal Angina:** There is a risk of exacerbating coronary artery spasm if patients with Prinzmetal or variant angina are treated with a beta-blocker. If this treatment is essential, it should only be undertaken in a Coronary or Intensive Care Unit.

9. **Euthyroid Hyperthyroxinaemia:** The effects of beta-blockers on thyroid hormone metabolism may result in elevations of serum free thyroxine (T₄) levels. In the absence of any signs or symptoms of hyperthyroidism, additional investigation is necessary before a diagnosis of thyrotoxicosis can be made.

Precautions:

1. **Anaesthesia and the Peri-operative Period:** Beta-blockade may have beneficial effects in decreasing the incidence of arrhythmias and myocardial ischaemia during anaesthesia and the post-operative period. It is currently recommended that maintenance beta-blockade be continued peri-operatively. The anaesthetist must be made aware of beta-blockade because of the potential for interactions with other drugs, resulting in severe bradyarrhythmias and hypotension, the decreased reflex ability to compensate for blood loss, hypovolaemia and regional sympathetic blockade, and the increased propensity for vagal-induced bradycardia. Incidents of protracted severe hypotension or difficulty restoring normal cardiac rhythm during anaesthesia have been reported. Modern inhalational anaesthetic agents are generally well tolerated, although older agents (ether, cyclopropane, methoxyflurane, trichlorethylene) were sometimes associated with severe circulatory depression in the presence of beta-blockade.

2. **Diabetes:** Beta-blockers affect glucose metabolism and may mask some important premonitory signs of acute hypoglycaemia, such as tachycardia. In patients with insulin or non-insulin dependent diabetes, especially labile diabetes, or with a history of spontaneous hypoglycaemia, beta-blockade may result in the loss of diabetic control and delayed recovery from hypoglycaemia. The dose of insulin or oral hypoglycaemic agent may need adjustment.

3. **Other Metabolic Effects:** Beta-adrenoreceptors are involved in the regulation of lipid as well as carbohydrate metabolism. Some drugs affect the lipid profile adversely although the long-term clinical significance of this

change is unknown and the effect appears to be less for drugs with intrinsic sympathomimetic activity.

4. **Use of Catecholamine-Depleting Agents:** Concomitant use of drugs such as reserpine and guanethidine requires careful monitoring since the added effect of a beta-blockade may produce an excessive reduction of the resting sympathetic nervous tone.

5. **Phaeochromocytoma:** In patients with this condition, an alpha-blocking drug (e.g. phentolamine/phenoxybenzamine) should be administered before the beta-blocker to avoid exacerbation of hypertension.

6. **Eye and Skin Reactions:** Various skin rashes and conjunctival xerosis have been reported with beta-blockers. Cross-reactions may occur between beta-blockers, therefore, substitutions within the group may not necessarily preclude occurrence of symptoms. During the long-term treatment with the beta-blocking drug, practolol, a specific rash bearing a superficial resemblance to psoriasis was occasionally described. In a number of patients affected, this rash was accompanied by adverse effects on the eye (xerophthalmia and/or keratoconjunctivitis) of varying severity. This condition is called the oculomucocutaneous syndrome, or practolol syndrome. In a few patients, these eye changes occurred independently of a skin rash. On rare occasions, serous otitis media, sclerosing peritonitis, pericarditis and pleurisy have been reported. Although the practolol syndrome has not been observed in patients taking other beta-blockers, the possibility of such side effects occurring should be borne in mind. More recently an association between Peyronie's disease (a fibrosing induration of the penis) and various beta-blockers has been suggested but is not proven.

7. **Allergic Conditions:** These may be exaggerated by beta-blockade (e.g. allergic rhinitis during the pollen season and allergic reactions to bee and wasp stings). Beta-blockers should be avoided if there is a risk of bronchospasm.

8. **Hyperthyroidism:** Because beta-blockade may mask the clinical signs of developing or continuing hyperthyroidism, resulting in symptomatic improvement without any change in thyroid hormone status, special care should be exercised in those patients who are hyperthyroid and are also receiving beta-blockers.

9. **Clonidine:** Concurrent use of beta-blockers and clonidine should be avoided because of the risk of adverse interaction and severe withdrawal symptoms. If administered concomitantly, the clonidine should not be discontinued until several days after the withdrawal of the beta-blocker.

10. **Renal Disease:** In patients with severe renal disease, haemodynamic changes following beta-blockade may impair renal function further. Beta-blockers which are excreted mainly by the kidney may require dose adjustment in patients with renal failure.

11. **Significant Cardiomegaly.**

12. Preparations containing lithium generally should not be given with diuretics because they may reduce its renal clearance.

13. **Hypokalaemia:** Hypokalaemia may occur during prolonged treatment with Pretenol-C. The risk of hypokalaemia is increased in the presence of a predisposition to potassium depletion, where electrolyte intake is restricted or during concomitant corticosteroid or ACTH administration. Predisposition may accompany hepatic or renal disease, hepatic cirrhosis, chronic diarrhoea or the use of purgatives. A potassium-rich diet (dried fruit, vegetables, fruit juices, fish, low-fat cheese) or a potassium supplement is indicated in these instances. Serum electrolyte levels should be monitored and signs of fluid / electrolyte imbalance looked for, namely hyponatraemia, hypochloraeic alkalosis and hypokalaemia. Symptoms include thirst, dryness of the mouth, muscle cramps, lethargy, paraesthesia, drowsiness, tachycardia and restlessness.

14. **Hyperuricaemia** may occur in patients receiving chlorthalidone. It is usually asymptomatic and rarely leads to clinical gout except in patients with a history of gout, familiar predisposition to gout, or chronic renal failure.

INTERACTION WITH OTHER MEDICATIONS:

1. Verapamil, (see Warnings)
2. Care should be taken in prescribing a beta-adrenoreceptor blocking drug with anti-arrhythmic drugs. (see Warnings)
3. Insulin and oral hypoglycaemics.
4. Anaesthetics such as ether, chloroform and cyclopropane are contraindicated with Atenolol.
5. Catecholamine-depleting drugs: such as reserpine or guanethidine may interact.
6. Calcium antagonists with myocardial depressant effects.
7. Clonidine
8. Digitalis/digitalis glucosides and beta blockers are commonly used together although there have been reports of excessive bradycardia when beta-blockers are used to treat digitalis intoxication.
9. Concomitant use of sympathomimetic agents, e.g. adrenaline, may counteract the effects of beta-blockers.
10. Concomitant use of prostaglandin synthetase inhibiting drugs, e.g. ibuprofen and indomethacin may decrease the hypotensive effects of beta-blockers.
11. Lithium: Since diuretics raise blood lithium levels, the latter must be monitored in patients under lithium therapy who are taking Chlorthalidone at the same time. Where lithium has induced polyuria, diuretics may exert a paradoxical antidiuretic effect.
12. Curare derivatives and antihypertensive drugs: Diuretics potentiate the action of curare derivatives and antihypertensive drugs (e.g. guanethidine. Methyldopa, beta-blockers, vasodilators, calcium antagonists, ACE inhibitors).
13. Corticosteroids, ACTH, amphotericin, and carbenoxolone: The hypokalaemic effects of diuretics may be increased by corticosteroids, ACTH, amphotericin, and carbenoxolone.
14. Non-steroidal anti-inflammatory drugs: Concomitant administration of certain non-steroidal anti-inflammatory drugs (e.g. indomethacin) may weaken the diuretic and antihypertensive activity of diuretics, and there have been isolated reports of a deterioration in renal function in predisposed patients.
15. **Allopurinol:** Coadministration of thiazide diuretics may increase the incidence of hypersensitivity reactions to allopurinol.
16. **Amantadine:** Coadministration of thiazide diuretics may increase the risk of adverse effects from amantadine.
17. **Antineoplastic agents (e.g. cyclophosphamide, methotrexate):** Concomitant use of thiazide diuretics may reduce renal excretion of cytotoxic agents and enhance the myelosuppressive effects.
18. **Anticholinergics (e.g. atropine, biperiden):** The bioavailability of thiazide-type diuretics may be increased by anticholinergic agents, apparently due to a decrease in gastrointestinal motility and stomach-emptying rate.
19. **Cholestyramine:** Absorption of thiazide diuretics is decreased by cholestyramine. A decrease of the pharmacological effect may be expected.
20. **Vitamin D:** Use of thiazide diuretics may decrease urinary excretion of calcium, and coadministration of Vitamin D may potentiate the increase in serum calcium.
21. **Cyclosporin:** Concomitant treatment with diuretics may increase the risk of hyperuricaemia and gout-type complications.
22. **Calcium salts:** Concomitant use of thiazide-type diuretics may cause hypercalcaemia by increasing tubular calcium reabsorption.
23. **Diazoxide:** Thiazide diuretic may enhance the hyperglycaemic effect of diazoxide.

USE IN PREGNANCY & LACTATION: Use in Pregnancy: Beta-adrenergic blocking agents may cause bradycardia in the foetus and newborn infant. During the final part of pregnancy and parturition, these drugs should therefore only be given after weighing the needs of the mother against the risk to the foetus.

Atenolol crosses the placental barrier in pregnant women, and under steady-state conditions, maternal and foetal blood levels of Atenolol are approximately equal.

No studies have been performed on the use of Atenolol in the first trimester and the possibility of foetal injury cannot be excluded. Atenolol has been used under close supervision for the treatment of hypertension in the third trimester. Administration of Atenolol for longer periods to pregnant women in the management of mild to moderate hypertension has been associated with intra-uterine growth retardation. The use of Atenolol in women who are, or who may become pregnant, requires that the anticipated benefit be weighed against the possible risks, particularly in the first and second trimesters.

Atenolol has been shown to produce a dose-related increase in embryo/foetal resorptions in rats at doses equal to or greater than 50 mg/kg. Although similar effects were not seen in rabbits, the compound was not evaluated in rabbits at doses above 25 mg/kg.

Use in Lactation: There is significant accumulation of Atenolol in breast milk. Caution should be exercised when Atenolol is administered to nursing women and the infant should be regularly assessed for signs of beta-blockade.

SIDE EFFECTS: Atenolol: Adverse reactions reported in clinical trials of Atenolol are mainly attributable to pharmacological actions. The adverse reactions listed below have been observed in patients in clinical trials who have received dosages of about 100 mg per day. It is not possible to give percentage incidences for each reaction, but if all mild and transient reactions are included as well as more serious ones, up to 10% of patients may experience some form of adverse reactions.

More Common Reactions: Gastrointestinal: Dry mouth, gastrointestinal disturbances including indigestion, constipation. Nervous System: Fatigue, dizziness. Respiratory: Wheezing, bronchospasm.

Less Common Reactions: Biochemical Abnormalities: Increases in SGOT, blood urea and serum creatinine have been reported. Cardiovascular: Bradycardia, left ventricular insufficiency, postural hypotension which may be associated with syncope, intermittent claudication may occur if already present, Raynaud's phenomenon, cold extremities, deterioration in heart failure, heart block. **Dermatological:** Rash, alopecia, psoriasisiform skin reactions, exacerbation of psoriasis. Gastrointestinal: Diarrhoea. Elevations of transaminase levels have been seen infrequently, rare cases of hepatic toxicity including intrahepatic cholestasis have been reported. Genito-Urinary: Impotence. Musculoskeletal: Ataxia. Nervous System: Vivid dreams, nightmares, paraesthesias, tinnitus, vertigo, malaise, headache, insomnia, mood changes, confusion. Ocular: Dry eyes, visual disturbances. Psychiatric: Hallucinations, depression, psychoses. Respiratory: Asthma, dyspnoea, nasal congestion. **Haemopoietic:** Thrombocytopenia, purpura. An increase in ANA (Antinuclear Antibiotics) has been observed, however the clinical relevance of this is not clear. Serious or Life-Threatening Reactions: Myocardial insufficiency may require treatment with digitalis and diuretics. Bradycardia may respond to atropine. Bronchospasm may be reversed with a beta2-stimulant. Hypotension, if severe, may require use of a vasopressor.

Chlorthalidone:

Electrolytes and metabolic disorders: Very common: mainly at higher doses, hypokalaemia, hyperuricaemia, rise in blood lipids. Common: hyponatraemia, hypomagnesaemia, hyperglycaemia. Rare: hypercalcaemia, glycosuria, worsening of diabetic metabolic state, gout. Very rare: hypochloraeic alkalosis.

Skin: Common: urticaria, other forms of skin rash. Rare: photo sensitisation.

Liver: **Rare:** intrahepatic cholestasis, jaundice.

Cardiovascular system: **Common:** postural hypotension, which may be aggravated by alcohol, anaesthetics or sedatives. **Rare:** cardiac arrhythmias.

Central nervous system: **Common:** dizziness. **Rare:** paraesthesia, headache.

Gastrointestinal tract: **Common:** loss of appetite, minor gastrointestinal distress. **Rare:** mild nausea and vomiting, gastric pain, constipation, diarrhoea. **Very rare:** pancreatitis.

Blood: **Rare:** thrombocytopenia, leucopenia, agranulocytosis, eosinophilia.

Eye disorders: **Frequency 'not known':** Choroidal effusion, acute myopia, acute angle-closure glaucoma

Other: **Common:** impotence. **Rare:** disturbances of vision. **Very rare:** idiosyncratic pulmonary oedema (respiratory disorders), allergic interstitial nephritis, vasculitis.

SYMPTOMS AND TREATMENT OF OVERDOSE: Atenolol: Poisoning and Overdosage: Overdosage has not been reported with atenolol but in overdosage with other beta-blocking agents, severe bradycardia and hypotension are commonly found. Acute heart failure and bronchospasm may also occur. Excessively diuresis should be countered by maintaining normal fluid and electrolyte balance.

Management: Severe bradycardia: Atropine, 1-2 mg intravenously may be used to induce vagal blockade. If bradycardia persists IV isoprenaline (25 micrograms initially or oriprenaline) 0.5 mg given by slow IV injection, may be given. In refractory cases, the use of a cardiac pacemaker may be considered.

Hypotension: Severe hypotension should respond to a sympathomimetic amine such as noradrenaline. In refractory cases, the use of glucagons hydrochloride should be considered.

Bronchospasm: Therapy with a beta2 stimulant such as salbutamol or terbutaline or therapy with aminophylline may be considered.

Acute cardiac failure: Conventional therapy with digitalis, diuretics and oxygen should be instituted immediately. In refractory cases, the use of IV isoprenaline followed, if necessary, by glucagons hydrochloride or IV aminophylline should be considered.

Chlorthalidone: Signs and symptoms: In poisoning due to an overdosage the following signs and symptoms may occur: dizziness, nausea, somnolence, hypovolaemia, hypotension, and electrolyte disturbances associated with cardiac arrhythmias and muscle spasms. Management: Induction of vomiting or gastric lavage and administration of charcoal. Intravenous fluid and electrolyte replacement may be indicated.

STORAGE CONDITIONS: Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZE: Pretenol C 50 TABLET: Pack in 3x10's.

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SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599 Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA